

A How the spacer composite score is built, term by term

Real output for rice OsDEP1, edit at position 305, build 96e270bb. The five candidate spacers the tool returns at this locus, ranked. Every term that enters the ranking is shown with its own value, so a user can see why a candidate sits where it does instead of accepting a single number. OsDEP1 is one of the 5 of 26 benchmark loci that trigger both the poly-T and the ΔG penalty; at most loci neither fires. The Pol III +1 base is checked but scores zero, because Ma et al. (2015) measured no efficiency difference between regular and irregular starts.

Rank	Spacer	PAM	Nick→edit	GC%	Points contributed by each scoring term					Σ terms	Composite score (1-99 scale)	
					Nick-to-edit distance	Spacer GC	Seed-region GC	Poly-T penalty	ΔG structure penalty			
1	S10	AGG	5 nt	45%	+50	+20	+10	0	−15	80		65
2	S11	AGG	2 nt	45%	+35	+20	+10	0	−15	65		50
3	S8	TGG	24 nt	30%	+35	+7	+5	−25	0	22		22
4	S7	AGG	41 nt	25%	0	+2	+5	−25	0	−18 <i>floored →</i>		1
5	S12	TGG	8 nt *	40%	−80	+14	+10	0	−5	−56 <i>floored →</i>		1

adds points removes points term not triggered

deeper colour = larger term

Σ terms adds the four sequence terms; the ΔG penalty is then subtracted. The composite score is clamped to 1–99, so a spacer whose terms sum below 1 is reported at the floor rather than as a negative number — S7 (−18) and S12 (−56) are both floored to 1, shown in red. * S12’s edit lies upstream of its nick. The reverse-transcriptase template extends downstream only, so it cannot reach that edit; the distance term becomes −80 instead of a reward, and no combination of the other terms can recover it. That row is in the panel to show what disqualification looks like.